

Satellite Computing Network Construction: Optimal Computing Node Deployment in Multi-Layer LEO Mega-Constellations

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Abstract—Satellite computing networks leverage the placement of computing resources on low-Earth orbit (LEO) satellite communication network to bring computing power closer to users, enabling robust and scalable solutions for emerging applications such as Internet of thing, edge computing and real-time analysis. However, with the explosion of multi-layer LEO mega-constellations (MLMCs), how to use the least number of computing nodes to achieve computing power resource coverage? This paper proposes a computing node deployment algorithm for MLMCs to construct satellite computing network with the minimum number of computing node. Specifically, we analyze the existence conditions of optimal computing node deployment, and derive the expression of the relationship between the number of computing nodes needed on the LEO, the multi-layer satellite network structure and the number of accessible hops of computing nodes. The expression can determine the minimum number of computing nodes needed to be placed on the MLMCs to realize optimal computing node deployment under a certain number of accessible hops. Then, according to the limitation of optimal computing node deployment, a multi-layer satellite network computing node deployment algorithm is proposed to determine the position of the computing resource in MLMCs, which can form a stable computing structure in satellite network. Finally, through simulation, the influence of network scale and computing node deployment on the number of required computing nodes and the effectiveness of signaling delay is analyzed. The simulation results show that this method can achieve better delay with the least number of computing nodes, balance the number of computing nodes and delay, and meet the specific network requirements.

Index Terms—Satellite computing network, multi-layer LEO communication network, mega-constellations, computing node deployment.

I. INTRODUCTION

SATELLITE computing network (SCN) has emerged as a critical application paradigm because of its wide coverage, low cost and strong computing power [1]. Benefiting from its extensive coverage and flexible deployment, satellite

computing has garnered considerable attention across various scenarios, including remote sensing and vehicle-to-vehicle communication in remote regions [2]. In these scenarios, with the escalating demand for quality of service (QoS), user terminals, constrained by limited computing and storage capabilities, must offer high-reliability and low-latency services to computationally intensive applications [3]. Consequently, enabling user terminals to offload computationally intensive and latency-sensitive applications to powerful satellite computing emerges as a viable solution. With the rapid development of edge computing and artificial intelligence, self-driving technology, telemedicine, industrial Internet and many other scenarios have been initially applied in satellite networks [4]. Therefore, multi-layer low Earth orbit (LEO) mega-constellations (MLMCs) network is a research hot spot in the current intelligent computing field [5].

MLMCs networks generally include hundreds or even thousands of satellite nodes. The global LEO satellite constellation has entered the stage of intensive deployment [6]. Starlink of SpaceX program in the United States has launched more than 7,000 satellites, accounting for 72% of the total number of LEO satellites in the world; “GW” constellation of China plans to deploy nearly 13,000 LEO satellites, covering 500 km and 1200 km orbits [7]. These constellations provide a wide-area data transmission channel for the computing network through the characteristics of high throughput and low delay [8]. For example, the bandwidth of OneWeb satellite single beam can reach 250 MHz and support the throughput of 7 Tbit/s [9].

The current satellite computing network is to build an intelligent service network covering the whole world through the deep integration of satellite constellations and ground computing facilities (GCFs) [10]. However, due to political and geographical factors, GCF global deployment cannot be realized. In order to obtain computing services, some signaling needs to be forwarded to the nearest GCF through the inter-satellite link (ISLs), which will increase the propagation delay of signaling and services [11]. Meanwhile, a large number of signaling backhaul will lead to serious local network congestion around GCFs, which will aggravate the network overhead and further aggravate the signaling delay [12].

In order to solve the problems caused by the global inability to deploy GCFs and the increase of constellation scale, it is a more reasonable choice to put the computing resources on

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the satellite. The current research focus is on software defined network (SDN), which can reduce the consumption of on-board resources and improve the flexibility and scalability of the network by separating the control plane from the data plane [13], and to equip the satellite with SDN controller. Others include placing the next generating node B (gNB), fog service controller [14] and other payloads with computing resources and capability on the satellite to realize the space-based computing network. Therefore, the problem of computing node deployment is an important problem to be solved in establishing satellite computing network. Geostationary Earth orbit (GEO) and middle Earth orbit (MEO) satellites have been widely recognized by the academic circles as substitutes for ground station (GS) to place computing resources and capabilities [15]. It is one of the current research directions to establish satellite computing network through placing computing resources on GEO or MEO satellites. However, although GEO and MEO can effectively improve the coverage of computing network, due to the high track height, the excessive propagation delay of signaling will seriously affect the stability and continuity of services.

By placing computing resources on LEO, a hierarchical satellite computing network can be constructed, which can effectively reduce the backhaul of signaling and improve the efficiency of network. However, the increase in the scale of satellite constellations also makes it difficult and costly to deploy computing power in satellite networks. Take starlink for example, the manufacturing cost of a single LEO satellite is about \$250-\$500 thousand. If every satellite in the MLMCs network deploys strong computing resources, it will cause high cost and waste of resources. Therefore, under the condition of limited deployment of GCFs, *is it necessary for every satellite node to have strong computing power? Which nodes need to deploy computing power? Does an optimal computing node deployment exist? How to achieve coverage of satellite network with the least number of computing nodes?*

In this paper, we study how to deploy computing resources on LEOs, and build an optimal computing node deployment in MLMCs network scenario. Through the geometric and mathematical analysis of the multi-layer LEO satellite network topology, the existence conditions of the optimal computing node deployment are deduced, and the multi-layer satellite network computing node deployment algorithm is proposed to determine the satellite nodes where the computing resources is placed in the satellite network, so as to guide the design of SCN. The main contributions of this paper are summarized as the following:

- **Existence conditions of optimal computing node deployment in MLMCs network:** This paper analyzes the influence of the computing node deployment on satellite networks. The coverage relationship between layers of multi-layer LEO satellite network and the topological structure of satellite network are explored. On this basis, according to the relationship between the number of computing node NC and the structure and scale of satellite network, the number of satellites covered by computing node $V(J)$ and the number of accessible hops J , $NC = \frac{N \times M \times L}{V(J)}$, the number of

computing node needed for MLMCs network can be quickly determined.

- **Multi-layer satellite network computing node deployment algorithm:** Through the analysis of the characteristics of optimal computing node deployment, it is found that the position relationship of the computing node is similar to the perfect error correcting code in coding theory. The algorithm links the position coordinates of the computing node with the error correcting codes, and determines the position of the computing node in MLMCs by determining a group of error correcting codes. The position of the computing node can be quickly determined in MLMCs network and form a stable computing structure in satellite network.
- **Comprehensive performance analysis and insights:** The influence of MLMCs network scale and computing node deployment on the number of required computing nodes and the effectiveness of computing service is analyzed. We simulate and analyze different satellite network scales, accessible hops and computing node deployment methods in specific management scenarios. It is found that the method proposed in this paper effectively reduces the delay of signaling distribution, but there is a trade-off between the number of computing nodes needed and the delay. In practical application, the performance between the number of computing nodes and the delay can be balanced by designing appropriate accessible hop points according to the delay demand and network scale.

The rest of this paper is organized as follows. The II section reviews the related work in the literature. Then, in the III section, the network model and optimal computing node deployment are defined and explained. In the IV section, the existence conditions of optimal computing node deployment are analyzed, and the multi-layer satellite network computing node deployment algorithm is proposed. In the V section, the simulation results in common network management scenarios are given. Finally, in the VI section, we summarized our work.

II. RELATED WORK

The SCN is an important direction for the future development of MLMCs, and it faces many challenges. Due to the rapid movement of satellites in low Earth orbit, user equipment may frequently handover the satellites it accesses. When user equipment moves from one satellite mobile edge computing platform to another, how to ensure service continuity is a key issue. Because the satellite coverage area in SCN is relatively small and the population density is different, the traffic demand is unbalanced. With the movement of satellites, the traffic received from ground nodes will change continuously according to the user density in the footprint area, resulting in the unbalanced utilization of communication and computing resources between different satellites [16]. With the appearance of inter-satellite links, satellite cluster has become a new trend of satellite network development. Satellites equipped with computing and storage resources form a satellite edge computing cluster, which can realize the migration of computing resources to the edge of the network. With the support of inter-satellite link and mobile

edge computing, satellite edge computing cluster will become more efficient and intelligent. However, SCN's task scheduling decision-making factors are very diverse and flexible, which makes the task scheduling problem difficult to solve [17].

The research direction of this paper is the deployment of computing resources. Due to the limited deployment of GCF, the computing network architecture dependent on GCF cannot meet the service requirements of low latency and high bandwidth in the future SCN [18]. Therefore, how to put the computing resource on the satellite to realize more flexible, efficient and convenient use of computing resources and improve the efficiency of SCN is one of the research directions of satellite network. As for the design of computing node deployment, the recent literature mainly focuses on the placement of SDN controllers in satellite networks and satellite-ground integrated networks [19]. The main difference between these existing placement strategies is the optimal indicator and problem solution.

The related optimization problems are modeled and analyzed by undirected graph [20], [21], [22], [23], [24], [25]. Because they are NP-hard problems, there is no general algorithm to solve these problems [26]. For solving this problem, the author in [22] modeled the placement problem as a mixed integer linear programming problem. In order to reduce the time complexity of mixed integer linear programming model, the author used submodule optimization technology to generate a near-optimal solution in a time-efficient way. In [23], the problem is expressed as an integer nonlinear problem with the goal of minimizing the average service delay and the constraints of resources, virtualised network function deployment and slot traffic, and a time expansion-based decoupled greedy algorithm is designed. In [24], the author uses particle swarm optimization algorithm to determine the optimal position of the gateway, but heuristic algorithms have insufficient accuracy in large-scale networks and are difficult to find the optimal solution. In [25], the author puts forward the problem of node deployment by static layout and dynamic allocation. Firstly, the SDN controller is integrated into some fixed satellites by establishing a mixed integer programming model, and then the switches are dynamically allocated to the existing controllers according to the switch-controller delay and the traffic load of the controllers.

However, the above methods are very complicated in modeling the problem, and the time complexity of the proposed algorithm is also very high. The number of satellite nodes in the applied satellite network scene usually does not exceed 100. But there will be thousands of satellites in the large-scale satellite network in the future [27], and the existing methods are difficult to meet the current needs. Moreover, the above methods only model and analyze single-layer satellite networks, and cannot be applied to MLMCs networks.

In addition to establishing the optimization model, some existing efforts are devoted to deriving closed-form formulas with time complexity. In [28], the author gives a series of theorems based on the principle of error correction code under Lee distance, which makes the optimal resource placement in torus network fast. Applying the above theory, the computing resource is placed on MEO satellite for satellite management

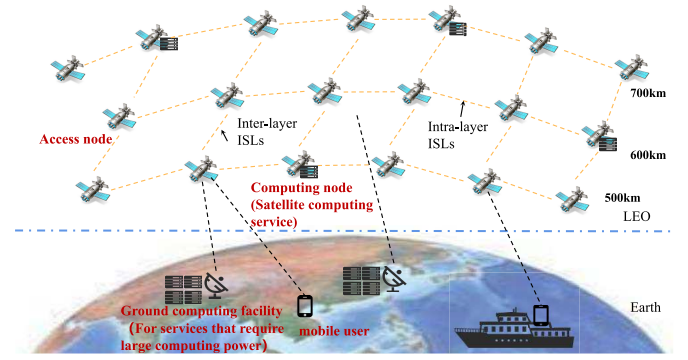


Fig. 1. Satellite computing network in MLMCs diagram.

in [29], a two-layer satellite network model is constructed, and the problem of constructing the optimal space control network is formulated from the perspective of satellite coverage factors, and the conditions for constructing the optimal space control network, space control network and satellite network configuration are derived by using geometric topology analysis. Although the method in [29] can be quickly applied to large-scale satellite networks, the model proposed in it can only be applied to the two-layer satellite network structure, and cannot be expanded in the number of layers, so it still has great limitations when applied to MLMCs networks.

Because of the high complexity, most optimization schemes are difficult to be quickly applied to practical engineering design, and the results of these works have not really reached the scale of MLMCs, and it is difficult to flexibly apply the description of satellite networks to multi-layer satellite networks, which is all the focus of this paper.

III. SYSTEM MODEL

In this section, the network model of MLMCs and the inter-layer beam coverage model of LEO satellites is introduced. Moreover, we analyzed the existence conditions of optimal computing node deployment for single-layer satellite. The key notations and abbreviations are shown in Table I.

A. Network Model

This paper considers a MLMCs network with computing empowerment. As shown in the Figure 1, we devolve some functions to LEO satellites to form a multi-level computing and management network architecture. All LEO satellites in the network assume the function of radio access network and provide services for users. Besides the function of accessing to the network, some satellites also undertake management functions that require computing resources.

1) *Network Architectures*: MLMCs is a coincidence network composed of $N_l \times M_l$ single-layer LEO Walker constellation with L layers height h_l kilometers (km), where N_l is the number of orbits at l -th layer and M_l is the number of satellites per orbit at l -th layer. LEO satellite set at l -th layer with an orbital height of h_l km is denoted by S_l , in which each satellite is denoted as a unique identifier $s_{l,n,m}$. The set of all LEO satellites is S . The layers communicate through

TABLE I
THE KEY NOTATIONS AND ABBREVIATIONS

Abbreviations/Symbols	Full name/Definition	Abbreviations/Symbols	Full name/Definition
LEO	low-Earth orbit	SDN	software defined network
MLMCs	multi-layer LEO mega-constellations	gNB	next generating node B
SCN	satellite computing network	GEO	geostationary Earth orbit
QoS	quality of service	MEO	middle Earth orbit
GCFs	ground computing facilities	GS	ground station
ISL	inter-satellite link	θ	beam coverage angle
N_{retx}	average transmission times	GCD	the greatest common factor
J	accessible hops	$d_{Lee}(x, y)$	Lee distance of vectors x and y
NC	number of computing node	$s(n, m, l)$	Satellite with number (n, m, l)
$V(J)$	number of satellites covered by computing node	SC_i	set of satellites covered by satellite i at the edge by one hop
$CT_q(t)$	set of all LEO satellites that computing node q can cover at time t	SB_i	the set of satellite covered satellite i by one hop beam
N, M, L	orbit number, number of satellites per orbit and layer number	P_t, G_t, G_r	transmission power, the gain of the transmitted and received antennas

beam coverage, and can only communicate with satellites in adjacent layers. The specific model will be discussed later. The access node inherits the functions of generation NodeB (gNB). Computing nodes can not only undertake the computing tasks of ground users, but also process on-board data, such as dealing with errors and faults in network operation, supporting real-time data processing of deep space exploration tasks and screening key information that needs to be transmitted to the ground, which can reduce backhaul and dependence on ground facilities [30]. Ground computing facilities are responsible for handling tasks that cannot be carried by on-board computing power.

2) *Computing Node*: The focus of this paper is to place the computing node on LEO. Therefore, unless otherwise specified, the computing nodes mentioned below are all computing nodes on LEO. Assume that the number of computing node nodes in the computing node set Q is $NC = |Q|$, where $|\cdot|$ represents the number of elements in the set. For each computing node $q \in Q$, the identifier vector and set of all LEO satellites that it can cover at time t are represented by $CT_q(t)$, which is called coverage domain. Considering the balance of satellite storage and processing load and signaling overhead, the computing node needs to cover the same number of LEO satellites $V = |CT_q(t)|$.

3) *Inter-Satellite Link*: Laser space stations between satellites in the same or adjacent orbital plane are created by laser transceivers of some existing satellite systems, among which in-orbit/adjacent orbital plane laser space stations are usually permanent [31]. We model the single LEO layer as a torus grid network. For the purpose of lucidity, this paper leverages a lattice depiction to encapsulate a two-dimensional (2D) torus. In this graphical representation, the enveloping edges are tangibly existent albeit tacitly omitted [23].

4) *Mobility*: To ensure the constancy of computing node load and computing coverage structure, LEO satellites should establish connections with computing nodes at the point of initial temporal instantiation or subsequent to a handover event. The connection process strictly adheres to the mapping relationship underpinning nonrepetitive coverage, which also helps to reduce the physical unreal of the torus lattice model. Because the inter-satellite links across layers are intermittent,

LEO satellites are equipped with electronically controllable radio beams to keep an eye on the fixed-covered LEO satellites to ensure a topologically stable structure from connection to the next handover, similar to the fixed coverage area of the earth.

B. Cross-Layer Coverage Model

The coverage areas of beams and satellites are significantly different in shape and size. For example, the starlink and Telesat use circular beams to form circular satellite coverage, in which the beam coverage area of the starlink is about 2,800 km², and the adjustable beam coverage area of Telesat is 960–246,000 km². The difference is that OneWeb has fixed and highly elliptical beams, each with a size of about 75,000 km², and the whole beam forms an approximately rectangular satellite coverage area [32]. In fact, beamforming technology can form any desired shape and size of satellite coverage. Although the establishment of cross-layer links between satellites is not stable at present, there have been many studies on cross-layer links [33]. Therefore, we assume that satellites can establish communication links with satellites in adjacent layers.

However, due to the relatively dense distribution of orbital heights at each layer of MLMCs network, satellite beams need high beam angles when covering multiple satellite nodes at adjacent layers. Because the satellite nodes are scattered in space and their relative positions are relatively fixed, the actual shape of the LEO satellite beam hardly affects the number of satellites covered. Therefore, this paper takes the shape of the adjacent satellite nodes actually covered by the satellite beam as the satellite beam. As shown in Figure 2(a), when the satellite beam can only cover one satellite, we call it point beam coverage. When the satellite beam can cover two satellites in the k -th layer, the minimum required beam coverage angle is as in (1), shown at the bottom of the next page. When the shape of satellite beam coverage is the same as the shape of one hop in the same layer, the minimum beam coverage angle required is as in (2), shown at the bottom of the next page. In Eq. 1 and Eq. 2, R is the Earth radius, h_l and h_{k+1} represent the orbital altitude of the k -th layer and

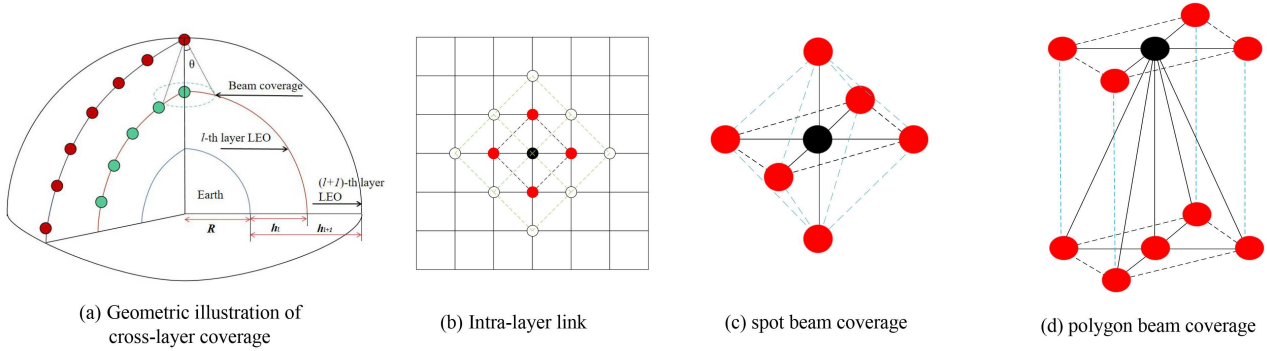


Fig. 2. MLMCs diagram: (a) represents the geometric illustration of cross-layer coverage (b) represents the intra-layer links; (c) represents the computing coverage structure of spot beam coverage; (d) represents the computing coverage structure of polygon beam coverage; In (b), (c) and (d), the black dot represents the computing node, and the red dot represents the node reached by 1 hop from the computing node. The solid line indicates that there is an inter-satellite link between satellites, and the dotted line indicates the range of the computing coverage structure, The intersections of lattice depiction represent satellites and each layer of lattice depiction is connected up and down, left and right to form a 2D torus.

$k + 1$ -th layer satellites respectively. In this paper, in order to describe the network topology conveniently, we only consider the establishment of inter-satellite links between satellites and satellites in adjacent layers, without considering the coverage of satellite beams across multi-layer satellites.

C. Computing Node Deployment

The computing node deployment includes the position and number of computing nodes, the shape and size of computing coverage structure, which has a great influence on the network performance such as network delay and load. In this paper, the goal of the computing node deployment scheme is to achieve non-repetitive coverage of all access satellite nodes with the least number of computing nodes.

Definition 1 (Optimal computing node deployment): At any time t , any node in the network can reach the computing nodes within at most J hops, and each computing node covers the same range without overlap, which can be defined in mathematical form as:

$$\begin{aligned} \forall t, \cup_{q=1}^Q CT_q(t) &= \sum_{l=1}^L S_l \\ CT_1(t) \cap CT_2(t) \cap \dots \cap CT_Q(t) &= \emptyset \\ \forall q = 1, \dots, Q, |CT_q(t)| &= V. \end{aligned} \quad (3)$$

It is difficult to determine the fixed network topology because of the uncertainty of the inter-layer links in multi-layer LEO satellite networks. However, the inter-satellite links in the single-layer satellite network are relatively stable, and the network extension can be analyzed as a 2D-tour structure. Therefore, we first analyze the computing node deployment problem of single-layer LEO satellite network.

At present, there are some existing works for reference about the computing coverage structure of single-layer low satellite network. We can get one of the conclusions from [29]: The satellite network of $N \times M$ is limited to the maximum hop count of J and the number of nodes in the computing coverage structure of $V(J)$, and there is an optimal computing node deployment if and only if N , M and J satisfy

$$V(J) | GCD(N, M), \quad (4)$$

where GCD stands for the greatest common factor. The computing coverage structure as shown in Figure 2(b)

Therefore, in order to determine whether there is an optimal computing node deployment, it is necessary to determine the coverage shape of computing node 1 hop, so as to determine the number of nodes in the computing coverage structure when J hops. In this paper, satellite nodes establish inter-satellite

$$\begin{aligned} \theta &= \frac{1}{2} \arccos \\ &\left(\frac{R + h_l - (R + h_{l+1}) \cos \left(\frac{3\pi}{\min(N_l, M_l)} \right)}{\sqrt{(R + h_l)^2 + (R + h_{l+1})^2 - 2(R + h_l)(R + h_{l+1}) \cos \left(\frac{3\pi}{\min(N_l, M_l)} \right)}} \right). \end{aligned} \quad (1)$$

$$\begin{aligned} \theta &= \arccos \\ &\left(\frac{R + h_l - (R + h_{l+1}) \cos \left(\frac{3\pi}{\max(N_l, M_l)} \right)}{\sqrt{(R + h_l)^2 + (R + h_{l+1})^2 - 2(R + h_l)(R + h_{l+1}) \cos \left(\frac{3\pi}{\max(N_l, M_l)} \right)}} \right). \end{aligned} \quad (2)$$

links with the upper and lower satellites in the same orbit and the left and right satellites in adjacent orbits in the same layer, which are diamond-shaped. Through analysis, we can get

$$\begin{aligned} V_I(1) &= 5 \\ V_I(J) &= V_I(J-1) + 4J \\ &= 2J^2 + 2J + 1. \end{aligned} \quad (5)$$

So we can draw a conclusion. For a $N \times M$ single-layer LEO satellite network with four inter-satellite links for a satellite, the conditions for realizing the optimal computing node deployment scheme within J hops of any node is if and only if

$$\begin{aligned} N &= \hat{N}, M = \hat{M}, J = \hat{J} \\ V(\hat{J}) &|GCD(\hat{N}, \hat{M}) \end{aligned} \quad (6)$$

IV. SATELLITE NETWORK TOPOLOGY ANALYSIS AND COMPUTING NODE DEPLOYMENT ALGORITHM

In this section, we analyze the existence conditions of optimal computing node deployment under different inter-layer links in MLMCs, and give the computing node deployment method under a given multi-layer LEO satellite network.

A. Spot Beam Coverage

As mentioned system model, when the satellite beam can only cover one satellite in the adjacent layer, we call it point beam coverage. For multi-layer LEO networks, generally, the structure of each layer is different and the inter-layer links are unstable, so it is difficult to establish a unified multi-layer network topology model. Therefore, we first analyze a multi-layer network with the same structure and stable inter-layer links, and then extend it to a more general structure.

Suppose a (N, M, L) multi-layer LEO satellite network, where N, M and L represent the number of orbits, the number of satellites per orbit and the number of satellite layers respectively. The satellites in each layer have the same structure, and the satellites in each layer are connected by four inter-satellite links. If we assume that each satellite can establish an inter-layer link with a satellite in its adjacent layer, the satellite network topology is similar to the three-dimensional tour (3D tour) structure $T_{N,M,L}$. In this case, the computing coverage structure realized by the computing node in 1 hop is shown in Figure 2(c). The difference is that there is no one-hop satellite link between the top and bottom layers of the satellite, so that the satellite network cannot form a ring in the third dimension. But we can do some analysis with the 3D tour structure first. For the convenience of analysis, we use T_k represents a 3D tour whose three dimensions are all k , where $k = GCD(N, M, L)$. Because $T_{N,M,L}$ is copies of T_k . By separately replicating the optimal computing node deployment of a T_k with $\frac{N}{k}$, $\frac{M}{k}$ and $\frac{L}{k}$ copies along three dimensions, we obtain an optimal computing node deployment for the given torus [28]. In the given network, each satellite can be uniquely determined by its layer, orbit and position in orbit, and we can uniquely represent a satellite with a set

of vectors $s = (n, m, l)$. And the number of nodes in the computing coverage structure is:

$$\begin{aligned} V_{II}(J) &= 2 \times \sum_{i=0}^{J-1} V_I(i) + V_I(J) \\ &= \frac{4}{3}J^3 + 2J^2 + \frac{8}{3}J + 1. \end{aligned} \quad (7)$$

Let set S contain vector sets of all satellite nodes. When searching for the position of computing nodes, it is a standard that any node can reach a computing node within at most J hops. That is to say, we need to find a set of vectors in set S , so that the Lee distance between any vector in S and the given vector is J at the maximum. Lee distance is to evaluate the distance between two vectors $x(x_1, x_2, \dots, x_n)$, $y(y_1, y_2, \dots, y_n)$ of length n and k -ary, which is defined as:

$$d_{Lee}(x, y) = \sum_{i=1}^n \min(|x_i - y_i|, k - |x_i - y_i|). \quad (8)$$

When $k = 2$ or $k = 3$, Lee distance is equivalent to Hamming distance. Our goal is to have the least number of computing nodes under a limited number of accessible hops. Therefore, the problem is formulated as

$$\begin{aligned} \min \quad & N_q = |Q| \\ \text{s.t.} \quad & \min(d_{Lee}(x_{l,n,m}, Q)) \leq J \\ & X = \{x_{l,n,m} | x_{l,n,m} = 0 \text{ or } 1\} \\ & Q = \{s_{l,n,m} | x_{l,n,m} = 1, s_{l,n,m} \in S\} \\ & 1 \leq l \leq L, 1 \leq n \leq N, 1 \leq m \leq M. \end{aligned} \quad (9)$$

This problem is difficult to fast get an effective result by general optimization algorithm in MLMCs network with thousands of satellites [29]. So we use other methods to analyze this problem. The optimal computing node deployment we are looking for is similar to the perfect error correcting code in coding theory. We need to find the position of a group of computing nodes so that each satellite node can belong to a unique computing node within J hop, which is the same as that for a perfect J -error-correcting code. Assume that the codeword in the code space is cd_i and a set of perfect J -error-correcting codes is PCD . According to the properties of error-correcting codes, we can get

$$\forall cd_i, \min(d_{Lee}(cd_i, PCD)) \leq J. \quad (10)$$

Any codeword in the space can be corrected to a legal codeword with a distance not greater than J . If there is a scheme that satisfies Definition 1, then this scheme is the optimal solution of Eq. 9. And the perfect error correcting code in coding theory meets our definition. The satellite node is regarded as a codeword in the vector space, and the computing node position set is regarded as a set of error correcting codes, and a basic computing node position set is obtained by constructing a set of error correcting codes. Because the coverage range of each computing node is the same, the computing nodes are evenly distributed in the satellite network. After a group of basic computing node positions are determined, the computing node positions of the whole network can be obtained by translating the basic

computing node positions. By optimizing the deployment of computing nodes, the system guarantees that each satellite node is exclusively covered by one computing node, with a fixed coverage area. This design inherently prevents a single point of failure from causing a total system shutdown. In the event of a computing node failure, the satellite nodes it covers can be redirected to adjacent computing nodes via cross-layer links, thereby containing the scope of the failure. Node localization, grounded in error correction code theory, ensures an even distribution of computing nodes, mitigating the risk of localized overload.

Given a (N, M, L) MLMCs network. The satellites in each layer are connected by four inter-satellite links and cross-layer links of spot beam coverage are established only at computing nodes.

Theorem 1: The network has an optimal computing node deployment if and only if

$$\begin{aligned} J &= 1 \\ \text{mod}(k^3, V(J)) &= 0 \\ k &= \text{GCD}(N, M, L). \end{aligned} \quad (11)$$

The number of computing nodes NC required for the network is

$$\begin{aligned} NC &= \frac{N \times M \times L}{V(J)} \\ &= \frac{N \times M \times L}{\frac{4}{3}J^3 + 2J^2 + \frac{8}{3}J + 1} \end{aligned} \quad (12)$$

Proof: In the three-dimensional vector space, there is only Lee distance single error-correcting code, which means that there is an optimal computing node deployment only when the satellite node arrives at the computing node within $J = 1$ hop. A 3D tour $T_{N,M,L}$ is copies of T_k along the three dimensions. By replicating the optimal computing node deployment of T_k , we obtain an optimal computing node deployment for the given network. When realizing the optimal computing node deployment, the number of satellite network nodes k^3 must be divisible by the number of nodes $V(J)$ in the computing coverage structure, which is also true for the given network $T_{N,M,L}$, so that the relationship between k and $V(J)$ and the number of computing nodes can be obtained. $\square$

A set of 1-error-correcting codes can be obtained by constructing linear codes. In order to facilitate the calculation, the layer, orbit and in-orbit serial numbers of satellites are counted from 0 to realize the correspondence between codewords and satellite node positions. First, construct a set of basis vectors $[1, 2, 0]$, $[0, 1, 2]$. Then construct linear codes according to $\text{mod}(i \times [1, 2, 0] + j \times [0, 1, 2], 7)$, where 7 is obtained from the Equation 7: $V(1) = 7$, i, j can be any integer. From this, we can get $V(1)^2$ number of error correcting codes, which is the computing node position satisfying the optimal computing node deployment. Finally, according to the size of the MLCLs, the position of the computing node is expanded, and all the computing node deployment positions can be obtained. Let's analyze the influence of the satellite network when the third dimension can't form a tour under the condition of optimal computing node deployment mentioned above. Because $J=1$,

the computing node can only achieve cross-layer coverage for satellite nodes in adjacent layers. Therefore, there will be vacancies at the top and bottom of the satellite network. However, the vacant nodes can reach the computing node within 2 hops at most.

Corollary 1: In the optimal computing node deployment obtained by $T_{N,M,L}$ satellite network, satellite nodes can reach the computing node at the same level within 2 hops at most.

Proof: Select a satellite node $s(n, m, l)$, and assume that the computing node position closest to s is $c(n, m, l + 1)$. There are four satellite nodes in the same layer with a distance of 1 from s , and their positions are

$$\begin{aligned} s_1(n+1, m, l), \quad s_2(n-1, m, l), \\ s_3(n, m+1, l), \quad s_4(n, m-1, l). \end{aligned} \quad (13)$$

The computing nodes closest to these four nodes are not in the $l+1$ layer, because

$$\|s_i + (0, 0, 1) - c\| \leq 2, \quad (14)$$

they will overlap with the coverage range of c , which does not meet the conditions of optimal computing node deployment. And they cannot all be in $l-1$ layer, because their coverage ranges will overlap with each other. Therefore, at least one computing node closest to these four nodes is in the l layer, and the distance from the satellite node s is 2. $\square$

This shows that satellite nodes can reach the computing node within a relatively limited number of hops under this structure even if all the satellite cross-layer link fail. In fact, there are a large number of cross-layer link in the above structure that are not used. Only the cross-layer link needs to be realized on the satellite node where the computing node is placed, which can meet the requirements.

Algorithm 1 The Multi-Layer Satellite Network With Spot Beam Computing Node Deployment Algorithm

Require: L layers LEO satellite network, and the l -th layer of network structure is $N_l \times M_l$, maximum number of hops from satellite to computing node J .

- 1: Calculate $V(J)$ according to the Equation 7
 - 2: $N^* = \max(N_l)$, $M^* = \max(M_l)$. According to the maximum number of orbits and satellites per orbit, construct an (N^*, M^*, L) network.
 - 3: Construct two basis vectors $a_1 = [J, J+1, 0]$, $a_2 = [0, J, J+1]$.
 - 4: Construct a set of linear codes $cl = \text{mod}(i\vec{a}_1 + j\vec{a}_2, V(J))$ by basis vectors, where $i, j \in \{0, 1, \dots, V(J)-1\}$. The obtained linear code is the position of the computing node CL in the network.
 - 5: Copy and translate the obtained computing node position CL to fill the whole (N^*, M^*, L) network.
 - 6: According to the structure of each layer of network $N_l \times M_l$, delete the part where the computing node position exceeds the limit, and then place the computing node in the corresponding position.
-

Even if the requirements for satellite cross-layer links are reduced, the existence conditions of optimal computing node

deployment are still very strict. Conversely, not all optimal solutions to Eq. 9 necessarily meet the definition. When Eq. 11 is violated, no perfect error-correcting code exists within the code space, implying that all viable solutions fail to satisfy the definition. Yet, this does not entail that Eq. 9 lacks an optimal solution. In the MLMCs network, the network structure of each layer of satellites is usually different, so the ideal computing coverage structure usually does not exist. In order to quickly determine the deployment position of computing node in MLMCs network, we apply the method of constructing linear code to obtain the position of computing node in satellite networks with different structures at each layer as shown in Algorithm 1. When the network structure of each layer is the same and Equation 1 is satisfied, the obtained computing node position satisfies the optimal computing node deployment. The obtained positions of computing nodes are shown in the Figure 3.

Let's delve into the time complexity analysis of Algorithm 1. The complexity for calculating $V(J)$ stands at $O(1)$, while determining the maximum network size for an L layers network incurs a complexity of $O(L)$. Constructing base vectors and generating linear codes necessitates a nested loop to produce all possible combinations, resulting in a complexity of $O(V(J)^2)$. Copying and translating node positions, as well as replicating the generated linear code positions across the entire virtual network (N^*M^*L), entails a complexity of $O(N^*M^*L)$. Additionally, deleting nodes that exceed the actual layer dimensions and mapping the computational nodes to real satellites also contribute a complexity of $O(N^*M^*L)$. The dominant factor in the time complexity is

$$O(V(J)^2 + N^*M^*L). \quad (15)$$

However, in practical applications, J is typically small, leading to a further degradation of complexity to $O(N^*M^*L)$. This complexity increases linearly with the scale of the network. When compared to traditional integer linear programming, heuristic algorithms with exponential complexity, and polynomial-time complexity methods, Algorithm 1 demonstrates advantages in large-scale satellite networks.

Let's analyze the gap between the computing node position obtained by Algorithm 1 and the optimal computing node deployment when $J = 1$ and the network structure of each layer is different. Suppose there is a $L = 3$ layers $N \times M$ satellite network and satisfies Equation 1. The number of computing nodes required is $\frac{3NM}{7}$. If the scale of the second layer becomes smaller, it becomes $N^* \times M^*$, $N^* \leq N$ and $M^* \leq M$. In this case, the relative position of the computing node in the layer has not changed, but the corresponding relationship of the cross-layer links has changed due to the change of the satellite relative position between layers, and the coverage of the computing coverage structure to the cross-layer nodes may be overlapped and omitted. In this case, the required number of resource nodes is about $\lfloor \frac{2NM + N^*M^*}{7} \rfloor = \lfloor \frac{NM(\frac{N^*M^*}{NM} + 2)}{7} \rfloor$. Because the relative position of satellites in different layers has changed, it is impossible to completely determine the position relationship between computing nodes.

(1) In the best case, each computing node still corresponds to the vacancy of the adjacent layer, and there are only vacancies for the upper and lower layers due to the decrease of the second layer computing nodes. According to Corollary 1, these extra vacancies are at a distance of 2 from the computing node and the number is

$$\nabla n = \lfloor 2 \times \frac{NM - N^*M^*}{7} \rfloor = \lfloor \frac{2NM(1 - \frac{N^*M^*}{NM})}{7} \rfloor. \quad (16)$$

(2) In the worst case, the position of all computing nodes is shifted, and each computing node overlaps with the coverage of the adjacent computing nodes. This means that the computing node can't effectively manage the adjacent layer satellites, which is the same as the complete failure of the cross-layer link. In this case, the number of extra vacancies with a distance of 2 from the computing node is

$$\nabla n = \lfloor \frac{2NM}{7} + \frac{N^*M^*}{NM} \frac{NM}{7} \rfloor = \lfloor \frac{2NM(1 + \frac{N^*M^*}{NM})}{7} \rfloor. \quad (17)$$

As a comparison, in the scheme that cross-layer links are not considered and computing nodes are placed independently on each layer, when $J = 2$, according to the result of Formula 5, the required number of computing nodes is $\lfloor \frac{2NM + N^*M^*}{13} \rfloor = \lfloor \frac{NM(\frac{N^*M^*}{NM} + 2)}{13} \rfloor$. As shown in Figure 2(b), the number of satellite nodes with a distance of 2 from each computing node is 8. The total number of satellite nodes 2 hops away from the computing node is $\lfloor \frac{8NM(\frac{N^*M^*}{NM} + 2)}{13} \rfloor$. Regarding the case of $J > 1$ under spot beam coverage, we will analyze it in the simulation results in Section V.

B. Polygon Beam Coverage

The above discussion is that the cross-layer link of the computing node only connects one satellite. In the case that the cross-layer link can cover multiple satellites, we regard the beam coverage of the computing node as a polygon. The access nodes covered by the same layer of the computing node and the upper and lower layers form a polyhedral computing coverage structure. According to the knowledge of geometry, the polyhedron that can fill the whole space seamlessly is limited, and the columnar is an easy choice. In order to realize the columnar computing coverage structure, the beam coverage shape of the computing node should be the same as the one hop coverage shape of the computing node. In other words, the satellite beam should cover five satellites, as shown in Figure 2(d). And the beam coverage angle of the computing node is calculated according to Equation 2. Assume that the computing node covers the number of satellites in the adjacent layer to be $C(J)$ through beam coverage, and the number of satellites covered by the computing node through intra-layer links $V(J)$ is obtained by Equation 5. In the case where the computing coverage structure is columnar, $C(1) = V(1)$. Like the point beam analysis, suppose a (N, M, L) multi-layer LEO satellite network. The satellites in each layer have the same structure, and the satellites in each layer are connected by four inter-satellite links. Each satellite node can communicate with the satellites in the adjacent layer through the cross-layer link.

When $J = 2$, while keeping the beam shape of the single satellite unchanged, the beam coverage shape of the upper node to the lower node remains the same as the in-layer coverage shape.

Corollary 2: For L layers $N \times M$ LEO networks, under the condition that $V(J)|GCD(N, M)$ is satisfied, if the beam coverage shape of a single satellite is the same as the coverage shape $V(1)$ of a single satellite, then the multi-layer LEO network can also achieve the same optimal computing node deployment, and the overall computing coverage structure is a column shape with the single-layer structure as the base. When the maximum number of hops is J , the optimal computing node deployment of $J + 1$ layers can be realized at most.

Proof: The coverage of a single satellite to an adjacent layer is equal to the projection of its hop coverage on the adjacent layer. Assume that the computing node is placed on the 1 layer, and the number of satellite nodes on the l -th layer covered by the computing node is $|CT_q^l(J)| = V_l(J)$. The shape of the computing coverage structure $CT_q^{l+1}(J)$ depends on $CT_q^l(J)$, so the coverage shapes of the satellites in the $(l + 1)$ -th and l -th layers depend on the coverage of $J - 1$ hops satellite in the $(l - 1)$ -th layer. $\square$

(1) Assume that the coverage range of the computing node q is the same at l -th layer and $l + 1$ -th layer, that is

$$\begin{aligned} CT_q^{l+1}(J) &= CT_q^l(J) + (0, 0, 1), \\ |CT_q^l(J)| &= |CT_q^{l+1}(J)|. \end{aligned} \quad (18)$$

When J becomes $J + 1$, $4J$ satellites at the edge of coverage on l -layer extend outward by one hop. The set of satellites covered by satellite i at the edge by one hop is called SC_i , $|SC_i| = V(1)$, and the set of satellite covered by one hop beam is SB_i . Because the coverage of one hop of the satellite is the same as that of the beam,

$$SB_i = SC_i + (0, 0, 1), |SB_i| = C(1), V(1) = C(1). \quad (19)$$

In this case, the coverage on the l -th layer is

$$CT_q^l(J + 1) = (\cup_{i=1}^{4J} SC_i) \cup CT_q^l(J), \quad (20)$$

the coverage on the $l + 1$ -th layer is

$$CT_q^{l+1}(J + 1) = (\cup_{i=1}^{4J} SB_i) \cup CT_q^{l+1}(J), \quad (21)$$

According to the Equation 18 and 19, we can get

$$\begin{aligned} CT_q^{l+1}(J + 1) &= CT_q^l(J + 1) + (0, 0, 1), \\ |CT_q^{l+1}(J + 1)| &= |CT_q^l(J + 1)|. \end{aligned} \quad (22)$$

(2) When $l = 1$ and $J = 1$, $CT_q^2(1) = CT_q^1(1) + (0, 0, 1)$, then according to mathematical induction, the conclusion holds for any l and J .

As can be seen from Figure 2(b), there is a great overlap in the coverage of cross-layer links set on each satellite. In fact, we only need to establish cross-layer links on the computing node, and then realize the same computing coverage structure through intra-layer links. However, in this case, the computing node can only cover three layers of satellites at most. Because the column is very regular in space, we can cover the upper and lower layers by placing the computing node on one layer, so we only need to select some layers to place the computing

node and realize the optimal computing node deployment when Equation 4 is satisfied. The position of the computing node is similar to that in the single-layer satellite network. Only need to select the layer where the computing node is placed, and then realize the computing node deployment method of single-layer satellite network on the corresponding layer. Let the number of computing nodes placed in each layer of satellite network be $(x_1, x_2, \dots, x_L)$. To minimize the number of computing nodes, the coverage ranges of computing nodes will not overlap. According to the fact that the number of satellite nodes covered by computing nodes is equal to the total number of nodes in the satellite network, the following equation can be obtained.

$$\begin{aligned} V(J) \times \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \dots \\ 0 & 1 & 1 & 1 \\ \vdots & & \ddots & \\ & & 1 & 1 & 1 & 0 \\ \dots & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{pmatrix} \\ \times (x_1 \ x_2 \ \dots \ x_L) = \begin{pmatrix} NM \\ NM \\ \vdots \\ NM \end{pmatrix} \end{aligned} \quad (23)$$

According to the results obtained from the Equation 23, $x_l = 0$ means that the computing node is not placed at the l floor, otherwise, the computing node is placed at the l floor.

Example 2: Equation 23 does not always give a unique solution. Taking $L = 5$ as an example, the result of the solution is $x_1 = x_4$, $x_2 = x_5$, $x_3 = 0$. In this case, we can make $x_1 = 0$, and the final result is to place computing nodes on the second and fifth layers. The result obtained by Equation 23 is considered from the point of view that the coverage ranges of computing nodes do not overlap. If we put all the computing nodes placed in the fifth layer in the fourth layer, the computing coverage structure obtained is the same, but more cross-layer links are constructed.

When the structure of each layer of satellite network is different, because the topological structure in each layer of satellite network is unchanged, the relative position relationship of the computing node has not changed, which ensures that the division of space by the computing node remains unchanged and the computing coverage structure remains stable. When $J = 1$, the coverage of adjacent layers can always be achieved by adjusting the coverage angle. When $J > 1$, the coverage of the computing node needs to rely on the inter-satellite link of the adjacent layer, which will affect the hop count of the satellite node reaching the computing node, but because the space division of the computing node remains unchanged, the satellites of the adjacent layer can reach the computing node within a certain propagation delay. The specific algorithm for determining the position of the computing node is similar to Algorithm 1. Because the layer where the computing node is located is determined in advance, it is only necessary to determine the track and the position in the track of the computing node by constructing a

Algorithm 2 The Multi-Layer Satellite Network With Polygon Beam Computing Node Deployment Algorithm

- Require:** L layers LEO satellite network, and the l -th layer of network structure is $N_l \times M_l$, maximum number of hops from satellite to computing node J .
- 1: Calculate $V(J)$ according to the Equation 5
 - 2: $N^* = \max(N_l)$, $M^* = \max(M_l)$. According to the maximum number of orbits and satellites per orbit, construct an (N^*, M^*, L) network.
 - 3: Calculate the serial number of the layer where the computing node needs to be placed according to Equation 23.
 - 4: Construct a basis vector $a = [J, J + 1]$.
 - 5: Construct a set of linear codes $cl = \text{mod}(i\vec{a}, V(J))$ by basis vectors, where $i \in \{0, 1, \dots, V(J)-1\}$. The obtained linear code is the position of the computing node CL in the corresponding layer.
 - 6: Copy and translate the obtained computing node position CL to fill the whole (N^*, M^*, L) network.
 - 7: According to the structure of each layer of network $N_l \times M_l$, delete the part where the computing node position exceeds the limit, and then place the computing node in the corresponding position.
-

two-dimensional linear code. The specific method is shown in Algorithm 2.

Assume that the computing node is placed at the l -th layer which satisfies the conditions of single-layer optimal computing node deployment. We find that among the V nodes, J^2 nodes are reserved, and the computing coverage structure remains unchanged. In this case,

$$N_{l-1} = N_l, M_{l-1} = \frac{J^2}{V} M_l. \quad (24)$$

But the distribution of satellites in orbit is uneven. In order to distribute satellites evenly, we need to keep more satellites. When

$$N_{l-1} = N_l, M_{l-1} = \frac{J^2 + J}{V} M_l, \quad (25)$$

the satellites are evenly distributed.

In general, the point beam avoids the coverage mismatch that plagues the polygon beam and ensures the stable connection despite the topological asymmetry. Polygonal beam can realize the deployment of $J \geq 2$, and the NC is greatly reduced by cubic scaling. However, the geometric dislocation of polygonal beams, large height difference or irregular layer size will lead to the coverage gap and delay jitter of beam edges.

The determination of J should consider the trade-off between delay and deployment cost. When the value of J is large, although the number of computing nodes can be reduced, the increase of hop number will lead to the increase of signal delay, which will concentrate the workload on a few computing nodes and pose the risk of network congestion. The reduction of J can balance the load distribution, but it will increase the deployment cost. The optimization of J is not a general strategy, and it needs to be customized according to the specific network scale, service requirements and beam

capacity. Large-scale networks tend to choose higher J for cost-effectiveness, while delay-sensitive deployments require lower J .

V. MANAGEMENT SCENARIO USE CASE ANALYSIS AND RESULTS

In order to verify the time effectiveness of the proposed computing node deployment algorithm, we compare the delays of different computing node deployment in common satellite network management scenarios.

A. Energy Consumption and Delays of Management Scenarios

1) *Signaling Distribution*: Signaling distribution is a process that the computing node distributes signaling to its covered LEO satellites. This scenario is a crucial component of numerous network management processes, such as routing maintenance, location update, and resource scheduling decisions, and also a key factor affecting the performance of the whole management procedures. The delay of signaling distribution is the average propagation delay during all non-LEO satellite computing nodes distributing signaling to each covered LEO satellite.

2) *Energy Consumption*: The energy consumption of the satellite network mainly includes the daily standby of the satellite, processing tasks, receiving and sending signals by the antenna, etc. In this paper, because only the handover task is considered, and other complicated computing tasks are not considered, the energy consumption of satellite standby and computing tasks is set to be constant. This paper mainly considers the influence of computing node deployment on the energy consumed by signaling distribution. The loss of electromagnetic wave in natural space is:

$$10 \log \left(\frac{P_t}{P_r} \right) = 32.45 + 20 \log(r \times f) - G_t - G_r, \quad (26)$$

where P_t is the transmission power of the antenna. According to the parameters of general satellite antenna, the antenna sensitivity is set to -105dbm, the antenna gain G_t, G_r is set to 18, and the satellite communication frequency f is 30GHz in Ka band. With power control, the signals have the same power when they reach the receiving satellite. The energy consumption of a single communication is

$$E = N_{\text{retx}} E_t$$

$$SNR = 10 \log \left(\frac{P_r}{P_n} \right), \quad (27)$$

where N_{retx} is the average transmission times and P_n is the noise power, which obeys Gaussian distribution. In the simulation, when SNR is less than the threshold, retransmission is performed. Because the cross-layer link needs beamforming, the energy consumption increases by an additional 20% [34].

3) *Fault Handling*: Because satellites are exposed to the cosmic environment, various faults may occur. If the fault is not found and handled in time, it will lead to many problems such as data forwarding failure, continuous retransmission, improper decision, etc. SCN provides an on-board solution for

TABLE II
CONSTELLATION PARAMETERS

Constellation name	Orbit number	Number of satellites per orbit	Satellite number	Inclination angle (°)	Altitude (km)
Starlink	9	277	2493	42	335.9
	42	59	2478	48	340.8
	9	283	2547	53	345.6
GW-A59	16	30	480	85	590
	40	50	2000	50	600
	60	60	3600	55	508
OneWeb	18	40	720	87.9	1200

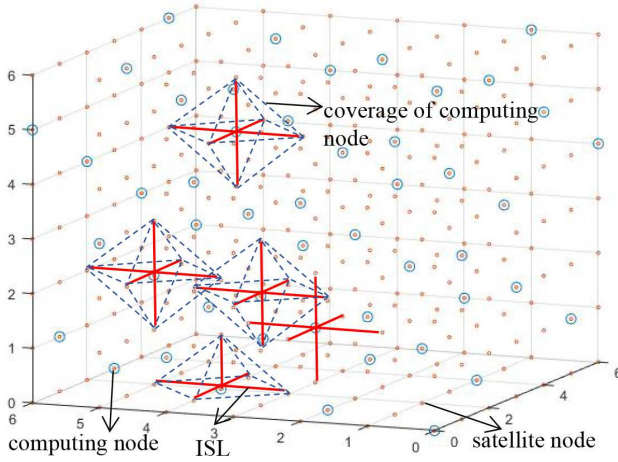


Fig. 3. Computing node deployment diagram: Blue represents the selected computing node, and red is the common satellite node. The red solid line represents ISL, and the blue dotted line represents the coverage of computing nodes. For the sake of visibility, not all ISL are marked.

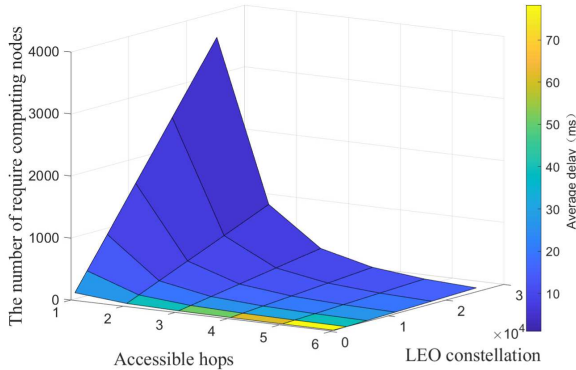


Fig. 4. The change trend of computing node number with accessible hop and LEO constellation scale and signaling distribution delay under spot beam coverage in $L = 7$ layers MLMCs network.

fault handling by integrating on-board computing power and dynamic resource scheduling. Usually, the satellite will report the status and problems to its computing node regularly, and then the computing node will sense the potential or existing faults, judge whether the faults can be solved on satellite and send solutions. The delay of fault handling includes the propagation delay of status report from LEO satellite to its computing node and the signaling distribution delay of fault information distribution.

B. Simulation Result Analysis

Set up an L layers $N \times M$ MLMCs network, the track height is evenly distributed in 500-1000km, and the computing can distribute signaling and make handover decisions. Through the algorithm in this paper, satellite nodes are selected to place computing nodes in the network, and simulation experiments are carried out by STK and Matlab. We reveal the influence of computing node deployment on time effectiveness and energy consumption of MLMCs network.

Figure 4 shows the changes of the number of computing nodes and the average delay of signaling allocation in MLMCs network with the increase of network size and accessible hop number when $L = 7$ and the beam coverage shape is plot beam. With the increase of accessible hops J , the number of computing nodes decreases rapidly, and then decreases steadily, which means that the gain of the number of computing nodes decreases gradually. Under the same number of accessible hops J , the number of computing nodes needed basically increases linearly with the satellite scale, but the larger the number of accessible hops J , the slower the growth rate. As comparisons, we select the three layers satellite in Starlink, GW-59 and OneWeb constellations as reference respectively. The specific parameters of each constellation are shown in Table II, and the number of computing nodes needed for different constellations is shown in Table III. On the other hand, because the increase of satellite scale reduces the distance between satellites, the average delay of signaling distribution decreases. Similarly, because the increase of accessible hop number increases the average distance between computing node and access satellite, the average delay increases.

Therefore, we have an interesting discovery that blindly improving the number of accessible hops will not promote the rapid and continuous decline of the optimal number of required computing nodes. Increasing the deployment of computing resources in the MLMCs network can also not quickly reduce the signaling distribution delay. The most important thing is how to balance the number of computing nodes and delay. In the design of computing resources placement, we can set appropriate accessible hops according to the delay demand and LEO constellation scale to balance.

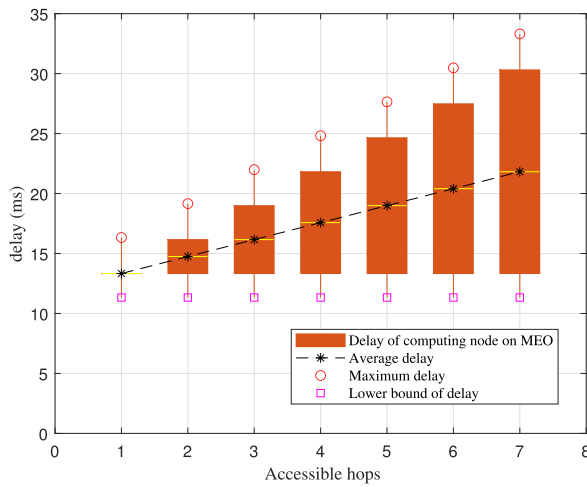
In Table IV, the number of compute nodes, the mean and standard deviation of the delay of different computing node deployment methods are compared. We changed the initial position relationship of computing nodes in this algorithm, and got a new set of computing node positions by keeping the number of computing nodes unchanged. In uniform

TABLE III
NUMBER OF COMPUTING NODES REQUIRED FOR DIFFERENT CONSTELLATIONS

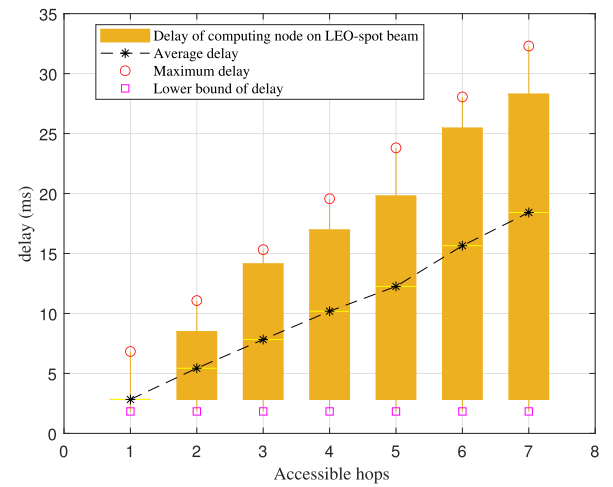
Constellation name	Satellite number	Accessible hop J						
		1	2	3	4	5	6	7
Starlink	7518	1503	578	300	183	123	88	66
GW-A59	6080	1216	467	243	148	99	71	53
OneWeb	720	144	55	28	17	11	8	6

TABLE IV
COMPARISON OF DIFFERENT COMPUTING NODE DEPLOYMENT ALGORITHMS

Computing node deployment method	Accessible hop J	1	2	3	4	5
Algorithm proposed in this paper	Signaling distribution delay/ms	2.42 ± 0.99	5.31 ± 2.17	7.80 ± 2.88	10.11 ± 3.70	11.49 ± 4.25
	Number of computing nodes	2500	700	277	135	75
Change the relative position	Signaling distribution delay/ms	3.23 ± 1.80	6.95 ± 3.20	13.28 ± 5.66	20.43 ± 10.54	29.48 ± 15.64
	Number of computing nodes	2500	700	277	135	75
Uniform distribution	Signaling distribution delay/ms	3.43 ± 1.91	5.68 ± 2.29	7.68 ± 2.77	9.54 ± 3.30	11.87 ± 3.82
	Number of computing nodes	2500	676	324	196	100



(a) Computing node placed on MEO



(b) Computing node placed on LEO-spot beam

Fig. 5. Signaling distributed delay of different computing node deployment.

distribution, the computing node positions are evenly selected in arithmetic progression manner according to the numbers, and the computing nodes are staggered between layers. The algorithm proposed in this paper can achieve better average signaling distribution delay with relatively few computing nodes.

The following compares the performance of different methods under a fixed satellite network scale. The network scale is $L = 7$ layers, $N = 50$, $M = 50$ satellite network. In the method of computing node on MEO, the orbital height of MEO is 5,000 km and beam coverage angle of MEO is 10° , assuming that MEO satellites can communicate with LEO satellites of all layers in the coverage area. As shown in Figure 5, the differences of signaling distribution delay of computing nodes in different positions are compared. The figure shows the changing trend of the maximum, minimum and average signaling distribution delay of the computing node on MEO and the computing node on LEO with the increase of the number of accessible hops of the computing node. With the increase of the accessible hops by the computing node, the maximum and average value of signaling distribution

delay increase linearly. Due to the closer physical distance to the access satellite, the signaling distribution delay of computing node on LEO satellite is significantly smaller than that of computing node on MEO. According to the analysis in Section III, the optimal deployment requires Equation 11. If this condition is not met, the nodes near the coverage edge will exceed the maximum number of hops. Therefore, with the increase of the number of accessible hops, the maximum signaling distribution delay of computing node placed on LEO increases faster. Considering the actual demand of satellite network, in order to avoid excessive delay, the number of accessible hops of the computing node should be controlled within a certain range.

Figure 6 shows the number of computing nodes required by different position of computing node under the given network scale. Since it is presumed that MEO satellites can communicate with satellites in every layer within their coverage area, MEO satellites possess superior communication capabilities compared to LEO satellites. Consequently, the computing nodes on MEO satellites can cover a greater number of satellites, necessitating fewer computing nodes. However, with

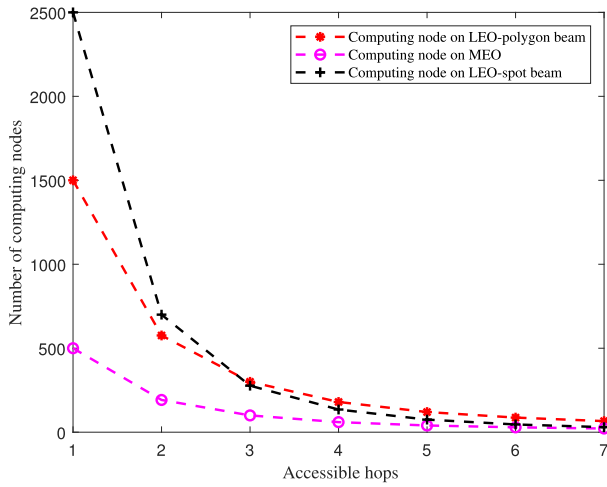


Fig. 6. Number of computing nodes required for different distribution.

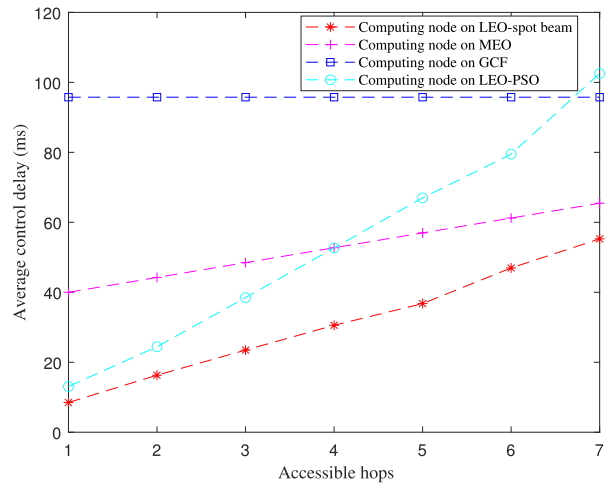


Fig. 7. Average delay of fault handling.

the increase of the number of accessible hops, the number of computing nodes required by different methods decreases rapidly at first, and then decreases steadily, and the number of computing nodes decreases faster when the computing nodes on LEO. Compared with Figure 5, it can be found that there is a trade-off among different schemes among accessible hops, number of computing nodes and signaling distribution delay. When choosing the computing node in the MLMCs network, we need to consider the actual needs of the network.

Figures 7 and 8 compare the network performance of different deployment methods. In order to control variables, the number of computing nodes deployed by particle swarm optimization (PSO) algorithm is the same as that of spot beam. The optimization objective is to minimize the average number of hops from satellite to computing node. Figure 7 compares the average delay of fault handling by computing nodes in different positions. When the fault is handled by GS, the satellite can communicate with GS through the inter-satellite link, and the result is the average time delay between the satellite in different positions and GS. For simple faults that can be processed on the satellite, the processing delay can be effectively reduced by the computing node on LEO

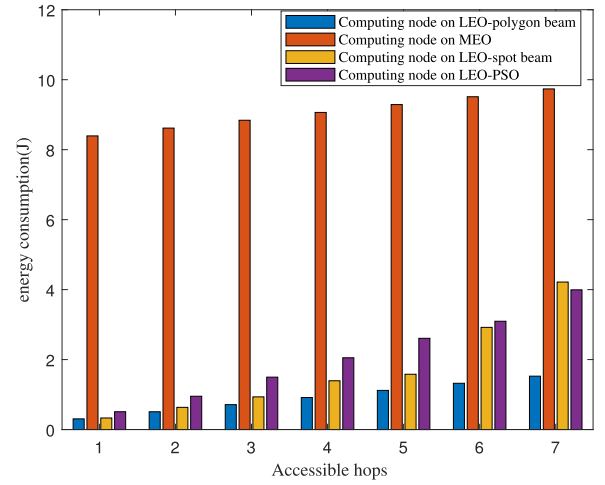


Fig. 8. Average energy consumption of different computing node deployment.

because the position of computing node is closer. Similarly placed on LEO, PSO algorithm needs to select a large number of computing nodes in the network, so it is difficult to achieve accurate optimization.

Figure 8 shows the change trend of the average energy consumed by signaling interaction with the increase of accessible hops when computing nodes in different locations distribute signaling. Because of the distance between MEO satellite and LEO satellite, signaling interaction requires satellite antenna to consume more energy, so the energy consumption is significantly higher than that of computing node on LEO. However, because there is no optimal computing node deployment when $J > 1$ in the spot beam mode, the average distance between the access node and the computing node increases, so the energy consumption increases faster. Also due to poor deployment, the energy consumption of PSO algorithm is higher. However, the handover service considered in this paper is relatively simple in energy consumption. In the future MLMCs network with computing empowerment, the satellite energy consumed in dealing with complex services may exceed the energy consumed by satellite antennas. Therefore, different schemes need to consider more complicated situations for the actual energy consumption.

VI. CONCLUSION

In this paper, an optimal computing node deployment is designed by putting the computing node on LEO satellite to build a satellite computing network in MLMCs. Firstly, we analyze the coverage mode between layers in the multi-layer LEO satellite network. Through the analysis of the topology of the multi-layer satellite network, we can quickly determine the conditions for building an optimal computing node deployment according to the structure and scale of the satellite network, the coverage characteristics of the computing nodes and the accessible hop count, and propose a multi-layer satellite network computing node deployment algorithm to select the satellite node where the computing node is located in the MLMCs network. Finally, the simulation results show that the method proposed in this paper improves the

time effectiveness of signaling distribution, and finds the relationship between the satellite network scale, the number of accessible hops of computing nodes, the number of computing nodes needed and the delay. The purpose of this work is to effectively guide the design of computing node deployment, and provide theoretical support for the design of satellite computing network in MLMCs.

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